The University of Texas at Austin
Cockrell School of Engineering

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BEYOND SMALL STRAIN: PROVING THE IMPORTANCE OF FINITE STRAIN THEORY

PGE 383 – Advanced Geomechanics
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Outline

- Introduction
  - Settlement of soil and Terzaghi's soil consolidation problem
  - Numerical model setup
  - Small vs large strains
- Methodology
  - FEM (Finite Element Methods)
  - Weak forms
  - MOOSE FEM solver architecture
- Results
  - Classical solutions
  - Bi-layer (Stratigraphy) model and solutions
- Discussion
- Conclusion
- References

INTRODUCTION

Settlement of Soil

- Total Settlement (S_t) = Immediate Settlement (S_i) + Primary Consolidation Settlement (S_c) + Secondary Consolidation Settlement (S_s)

Primary Consolidation

- Applied load causes the rearrangement of soil particles and dissipation of excess pore pressure in saturated soils.

Terzaghi's 1D Soil Consolidation Problem

- Theory focuses on expulsion of pore water from the voids leading to a decrease in volume and subsequent settlement.
- The one-dimensional consolidation process is governed by Terzaghi's consolidation equation:

$$\frac{\partial U}{\partial t} = C_v \frac{\partial^2 U}{\partial z^2} \quad \forall\, z \in (0, H), t > 0$$

where,

$U(z,t) = excess\ pore\ water\ pressure\ at\ depth\ z\ and\ time\ t$

$C_v = coefficient\ of\ consolidation$

$H = total\ thickness\ of\ the\ consolidating\ layer$

$t = time$

$z = vertical\ coordinate, measured\ downwards\ from\ the\ top\ surface\ (z = 0)\ to\ the\ bottom\ (z = H)$

TEXAS ENGINEERING

Problem Set up

- Single drainage (impermeable layer)
- Initial Condition

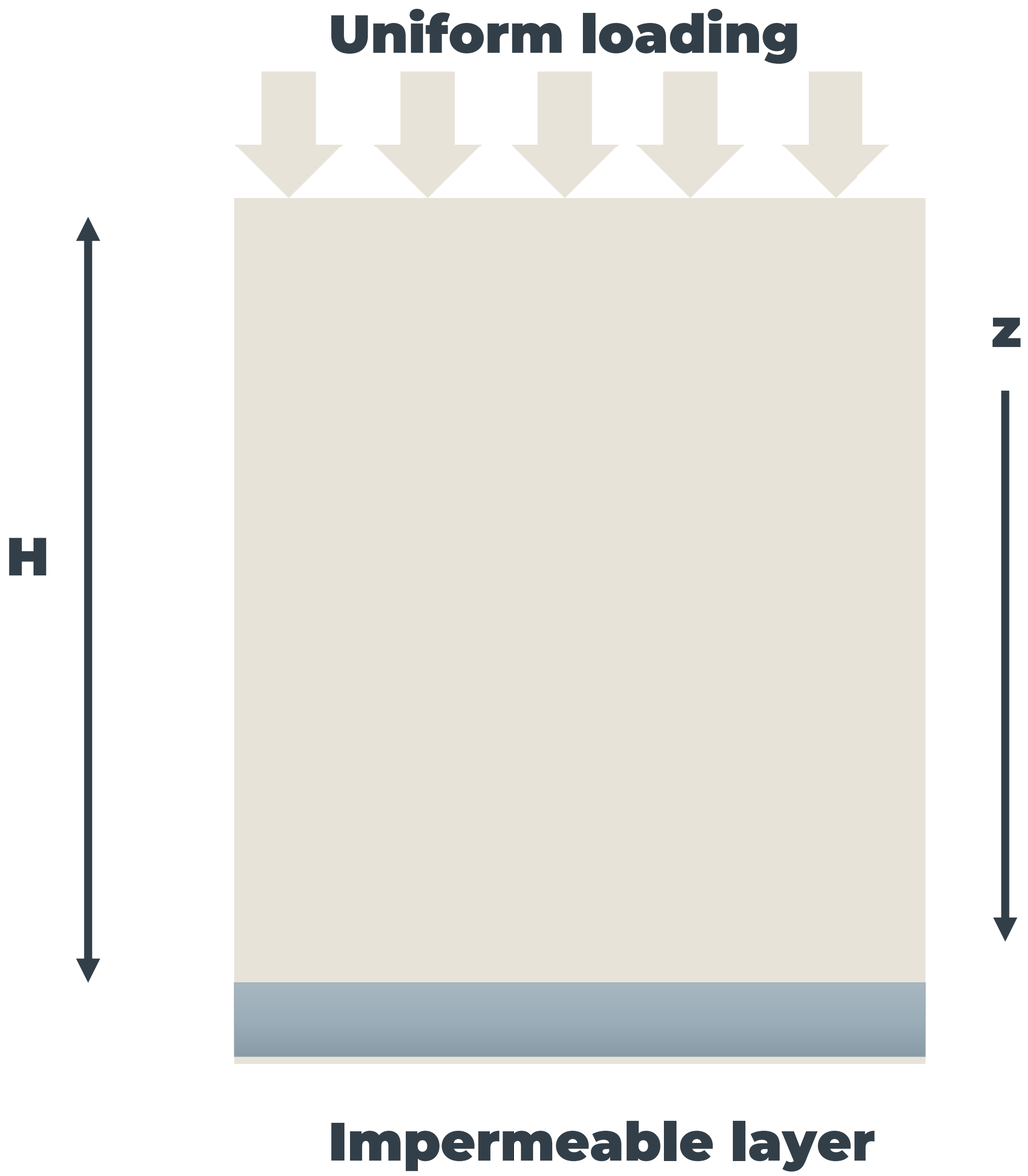
$$U(z,0) = f(z) for\ z \in [0,H]$$

- Boundary conditions

$$U(0,t) = 0$$

$$\frac{\partial U}{\partial z}(H,t) = 0$$

$$\sigma_{zz}(0,t) = q$$



Small Strain vs Large Strain

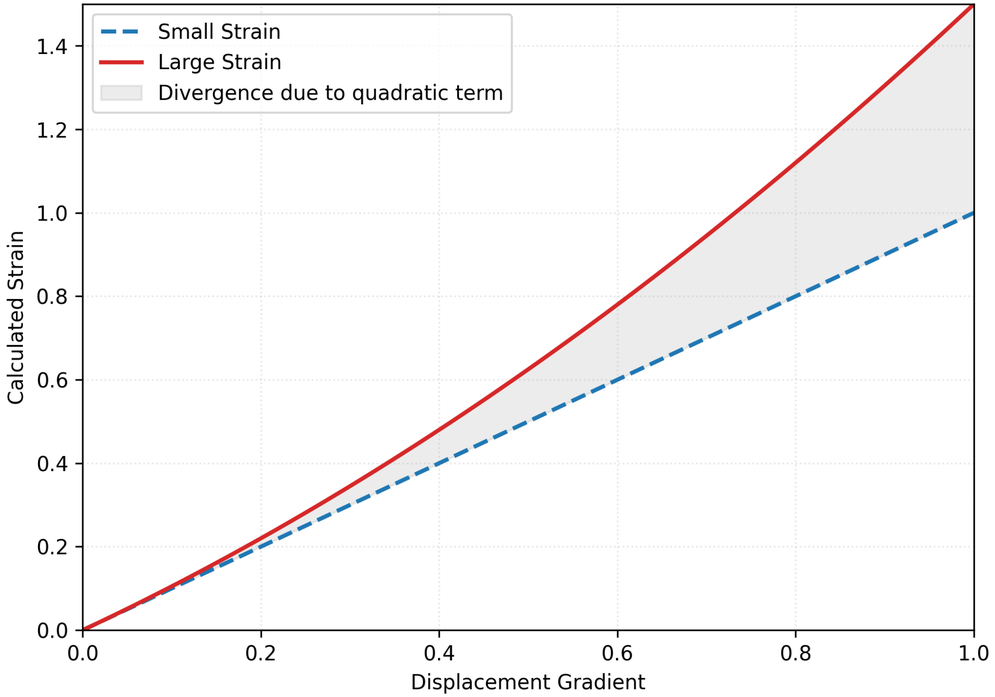
- Small strain - or small displacement - refers to the case where we assume that changes after a displacement is so small that the geometry is virtually unchanged.
- The key difference between large and small strain formulations is the exclusion of quadratic terms in strain calculations.
- The presence of quadratic terms in the large strain formulation makes the calculation nonlinear.

Cauchy (infinitesimal) strain,

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla_{\mathbf{X}}\mathbf{u} + (\nabla_{\mathbf{X}}\mathbf{u})^{\mathrm{T}})$$

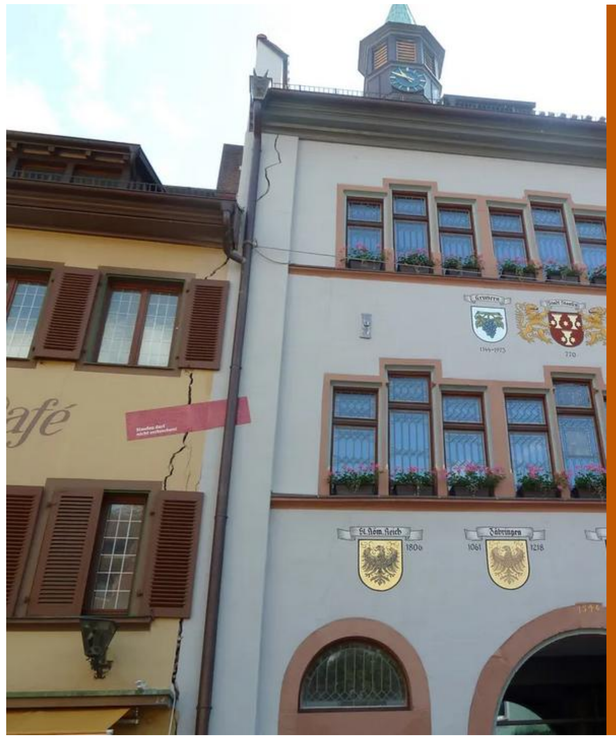
Green Lagrange Strain

$$\mathbf{E} = \frac{1}{2}((\nabla_{\mathbf{X}}\mathbf{u} + (\nabla_{\mathbf{X}}\mathbf{u})^{\mathrm{T}} + (\nabla_{\mathbf{X}}\mathbf{u})^{\mathrm{T}}\nabla_{\mathbf{X}}\mathbf{u})$$



Real World Consequences: The Staufen Disaster

- Trigger (2007) Geothermal drilling accidentally perforated an Anhydrite layer, connecting it to a groundwater aquifer.
- Reaction Water ingress triggered the hydration of Anhydrite into Gypsum, causing massive subsurface swelling.
- Damage
  - Ongoing Uplift: The ground rose by **>12 cm** (~ 1 cm/year)
  - Destruction: Cracks up to **15 cm (6 inches)** wide appeared in the historic town center.
  - Cost: **270+** buildings damaged with repair costs > **€50 Million**.



Structural damage in Staufen: Interior floor heave (left) and exterior fracturing (right) caused by anhydrite swelling (Sass & Burbaum, 2010; Ekem, 2011).

METHODOLOGY

From Strong Form to Weak Form in FEM

- The equations governing consolidation are given as partial differential equations (PDEs), which represent the strong form of the problem. To solve these using FEM (Finite Element Method) in MOOSE (FEM solver), we must convert them into weak form.
- Steps:
  - Multiply the strong form by a test function
  - Integrate over the entire domain (Ω)
  - Apply Integration by Parts (Divergence Theorem)

Weak Form: Small Strain

Equilibrium Equation:

Strong form:

$$\nabla \cdot \boldsymbol{\sigma}_{\text{total}} + \mathbf{b} = \mathbf{0}$$

In small deformation analysis,

$$\boldsymbol{\sigma} = \mathbf{D} : \boldsymbol{\varepsilon} \quad \textbf{or} \quad \boldsymbol{\sigma}_{ij} = \mathbf{D}_{\mathbf{ijkl}}\, \boldsymbol{\varepsilon}_{\mathbf{kl}}$$

Weak form

$$-\int_{\beta_0} \boldsymbol{\sigma}_{total} : \delta\boldsymbol{\varepsilon}\, \mathrm{dV} + \int_{\beta_0} \mathbf{b} \cdot \delta\mathbf{u}\, \mathrm{dV} + \int_{\partial\beta_0} \mathbf{T}_0 : \delta\mathbf{u}\, \mathrm{d}\mathbf{A} = 0$$

where,

$$\boldsymbol{\sigma}_{total} = total\ Cauchy\ stress\ tensor$$
$$b = body\ or\ intertial\ forces\ vector$$
$$\mathbf{T}_0 = \boldsymbol{\sigma}_{total} . \mathbf{n}$$
$$\delta\mathbf{u} = test\ function$$
$$\delta\boldsymbol{\varepsilon} = virtual\ small\ strain\ tensor$$

Weak Form: Small Strain

Mass Conservation Equation:

Strong form:

$$\alpha\, \nabla \cdot \dot{\mathbf{u}} \; + S_{\in}\, \dot{p} - \nabla \cdot \left(\frac{\mathbf{k}}{\mu}\, \nabla p \right) \; = \mathbf{0}$$

Weak form

$$\int_{\beta_0} \psi\, \alpha (\nabla \cdot \dot{\mathbf{u}}) dV + \int_{\beta_0} \psi S_{\in}\, \dot{p}\, dV + \int_{\beta_0} \nabla \psi \cdot \left(\frac{\boldsymbol{k}}{\mu}\, \nabla p \right) dV \; - \int_{\partial \beta_0} \psi\, \mathbf{Q}_n dA \; = 0$$

where,

$$\alpha = biot\ coefficient$$
$$k = intrinsic\ permeability$$
$$S_{\in} = constrained\ specific\ storage$$
$$\mu = fluid\ viscosity$$
$$p = pressure$$
$$\psi = test\ function$$
$$\mathbf{Q}_n = \mathrm{q}.\boldsymbol{n}$$

Weak Form: Large Strain

Equilibrium Equation:

Strong form:

$$\nabla_{\mathbf{X}} \cdot \mathbf{P} + \mathbf{b}_{\mathbf{X}} = \mathbf{0}$$

In finite deformation analysis,

$$\mathbf{E} = \frac{1}{2}\left(\mathbf{F}\,\mathbf{F}^{\mathrm{T}} - \mathbf{I}\right)$$

Weak form

$$-\int_{\beta_0} P : \delta\mathbf{F}\,\mathrm{dV} + \int_{\beta_0} \mathbf{b}_{\mathbf{X}} \cdot \delta\mathbf{u}\,\mathrm{dV} + \int_{\partial\beta_0} \mathbf{T}_0 : \delta\mathbf{u}\,\mathrm{d}\mathbf{A} = 0$$

where,

$$\mathbf{P} = First\ Piola - Kirchhoff\ (PK1)\ stress\ tensor$$
$$\mathbf{b}_{\mathbf{X}} = body\ or\ intertial\ forces\ vector\ per\ unit\ reference\ volume$$
$$\mathbf{T}_0 = \boldsymbol{P} \cdot \boldsymbol{N}$$
$$\delta\mathbf{u} = test\ function$$
$$\delta F = virtual\ deformation\ gradient\ (\delta\mathbf{F} = \nabla_{\mathbf{X}}(\delta\mathbf{u})$$

Weak Form: Large Strain

Mass Conservation Equation:

Strong form:

$$\alpha\,(\nabla_{\mathrm{x}} \cdot \mathbf{v}_{\boldsymbol{s}}) + \nabla_{\mathbf{x}} \cdot \mathbf{q} - \boldsymbol{S}_{\in}\,\dot{p} \;\; = \mathbf{0}$$

Weak form

$$\int_{\beta_0} \psi\,\alpha \dot{\boldsymbol{J}} dV + \int_{\beta_0} \psi \boldsymbol{J} S_{\in}\,\dot{p}\,dV + \int_{\beta_0} \nabla_{\mathbf{X}}\psi \cdot \left(\frac{\mathbf{K}}{\mu}\,\nabla_{\mathbf{X}} p\right) dV \; - \int_{\partial\beta_0} \psi\,\mathbf{Q}_n dA \; = 0$$

where,

$$\alpha = biot\ coefficient$$
$$J = Jacobian\ (local\ volume\ change\ ratio)$$
$$\mathbf{K} = pulled\ back, deformation\ dependent\ permeability$$
$$S_{\in} = constrained\ specific\ storage$$
$$\mu = fluid\ viscosity$$
$$p = pressure$$
$$\psi = test\ function$$
$$\mathbf{Q}_n = \mathbf{q}.\boldsymbol{n}$$

Analytical Solution of 1D Terzaghi Consolidation

The classical one-dimensional Terzaghi consolidation equation for excess pore water pressure $u(z,t)$ is,

$$\frac{\partial U}{\partial t} = C_v \frac{\partial^2 U}{\partial z^2} \quad \forall\, z \in (0,H), t > 0$$

where, C_v is the coefficient of consolidation.

Analytical solution for excess pore water pressure is,

$$u(z,t) = \sum_{m=0}^{\infty} \frac{2u_0}{M} \sin\left(\frac{M_z}{H}\right) \exp(-M^2 T_v)$$

where, $M = \frac{\pi}{2}(2m+1)$

Analytical solution for the degree of consolidation (U_{avg}) over time factor (T_v),

$$U_{avg} = 1 - \sum_{m=0}^{\infty} \frac{2}{M^2} \exp(-M^2 T_v)$$

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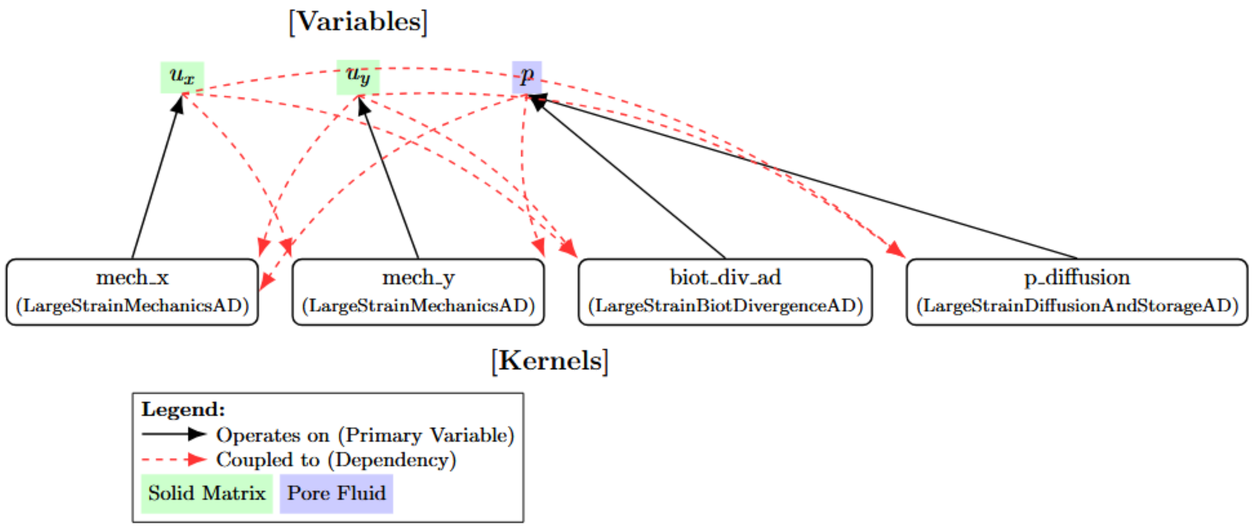
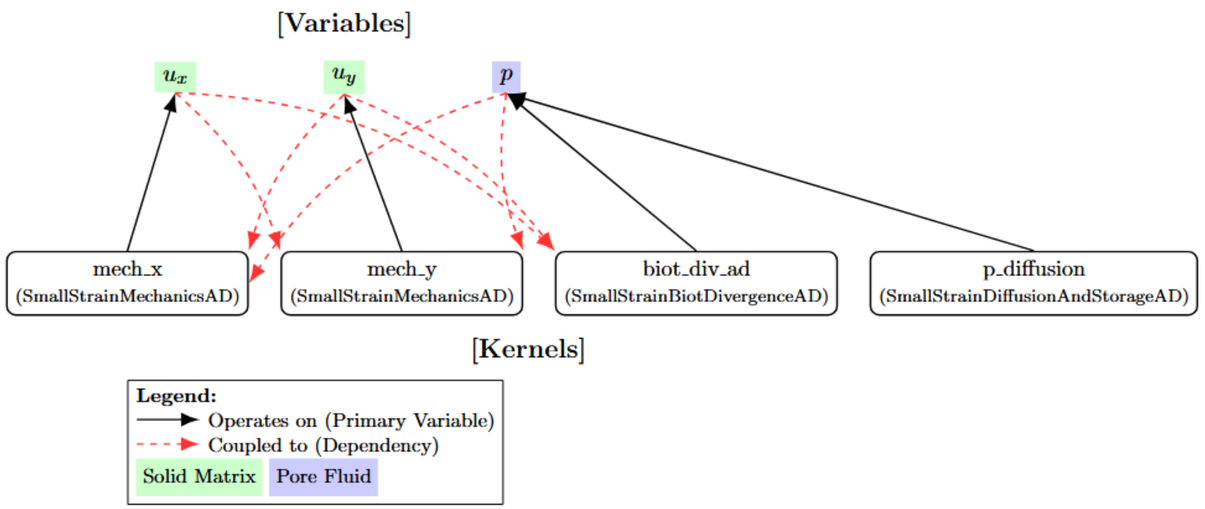
MOOSE Architecture

Large-Strain Consolidation Network

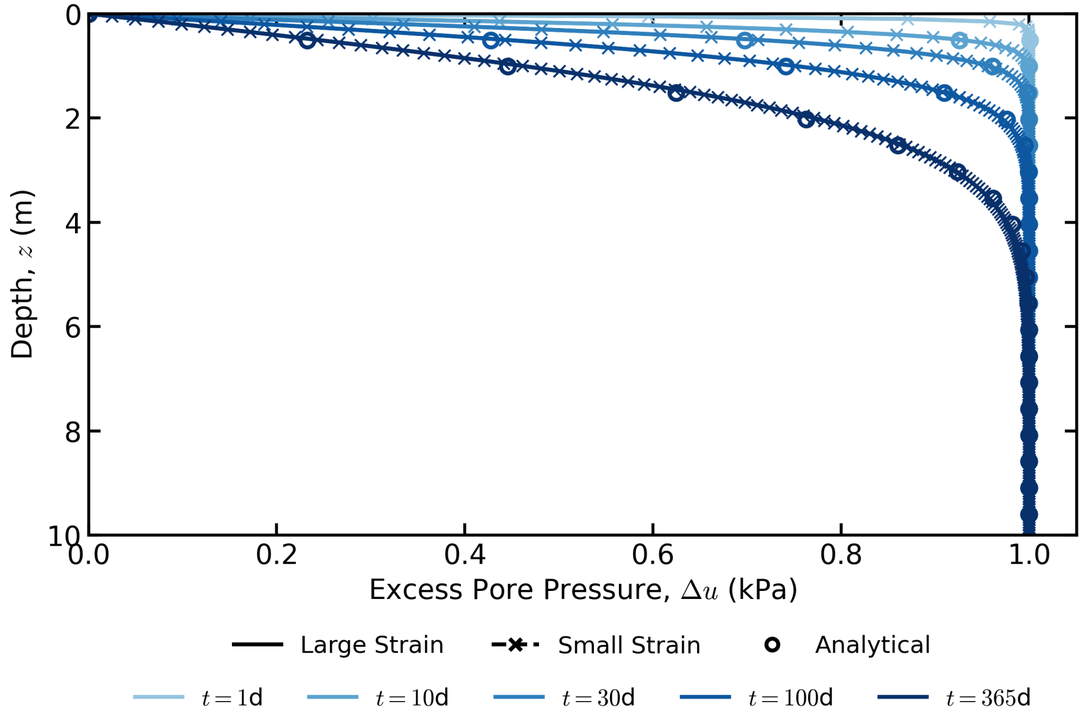


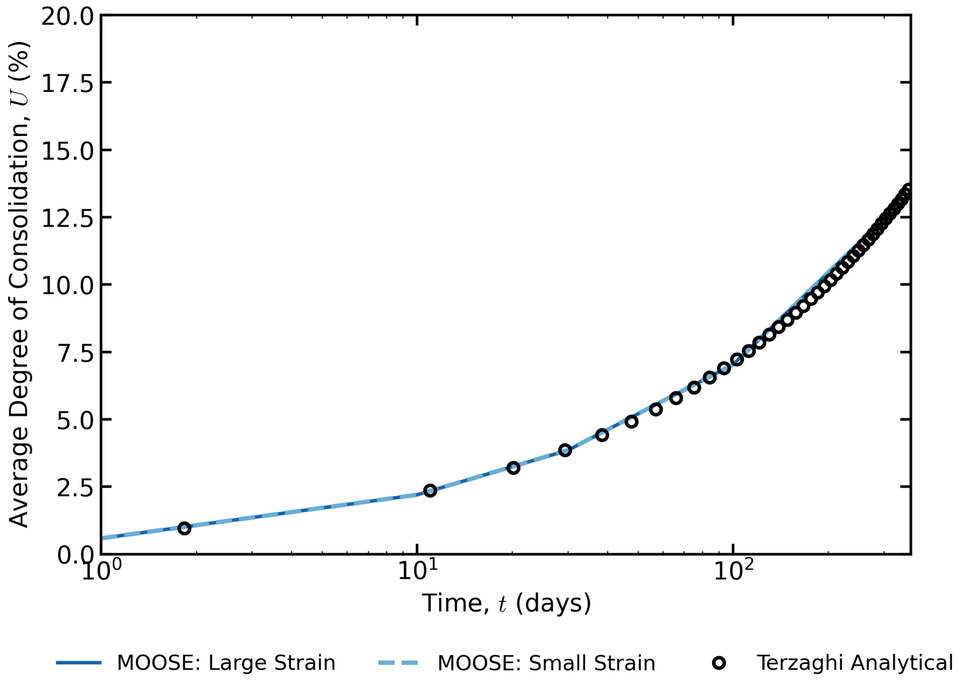
Small-Strain Consolidation Network



RESULTS

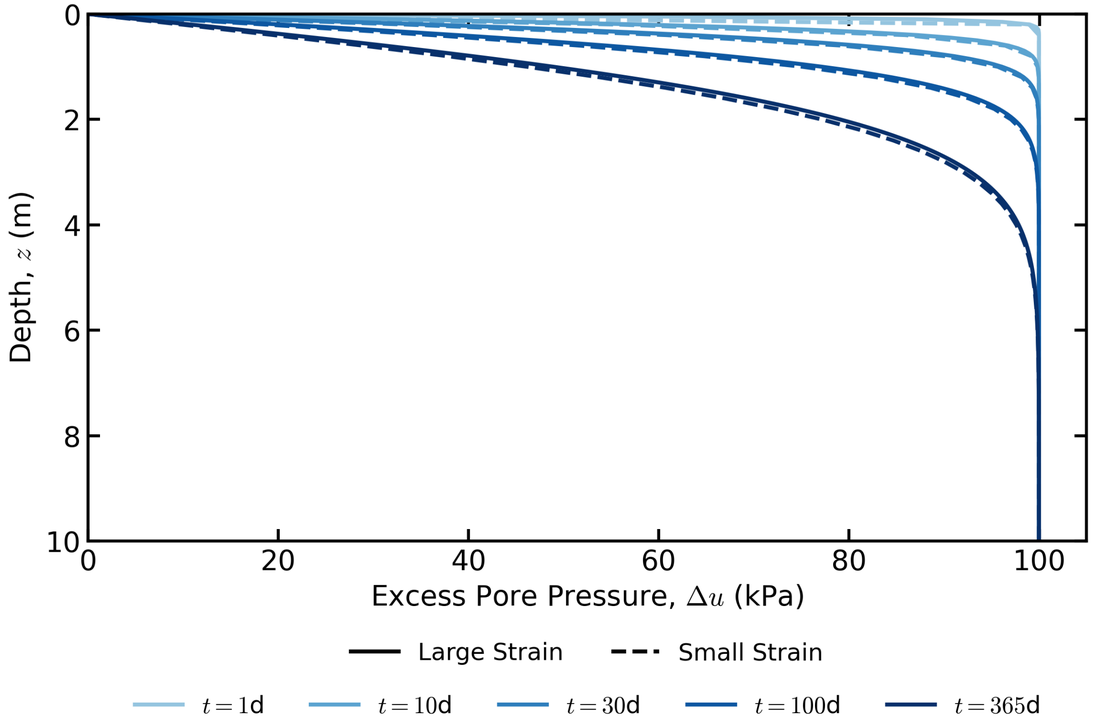
Results for Classical Model (Load = 1 kPa)

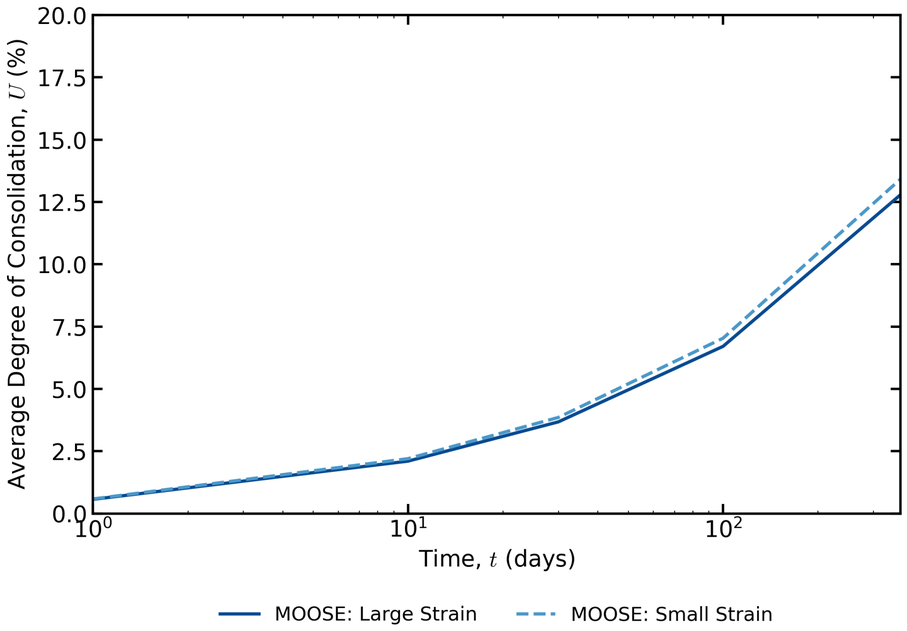




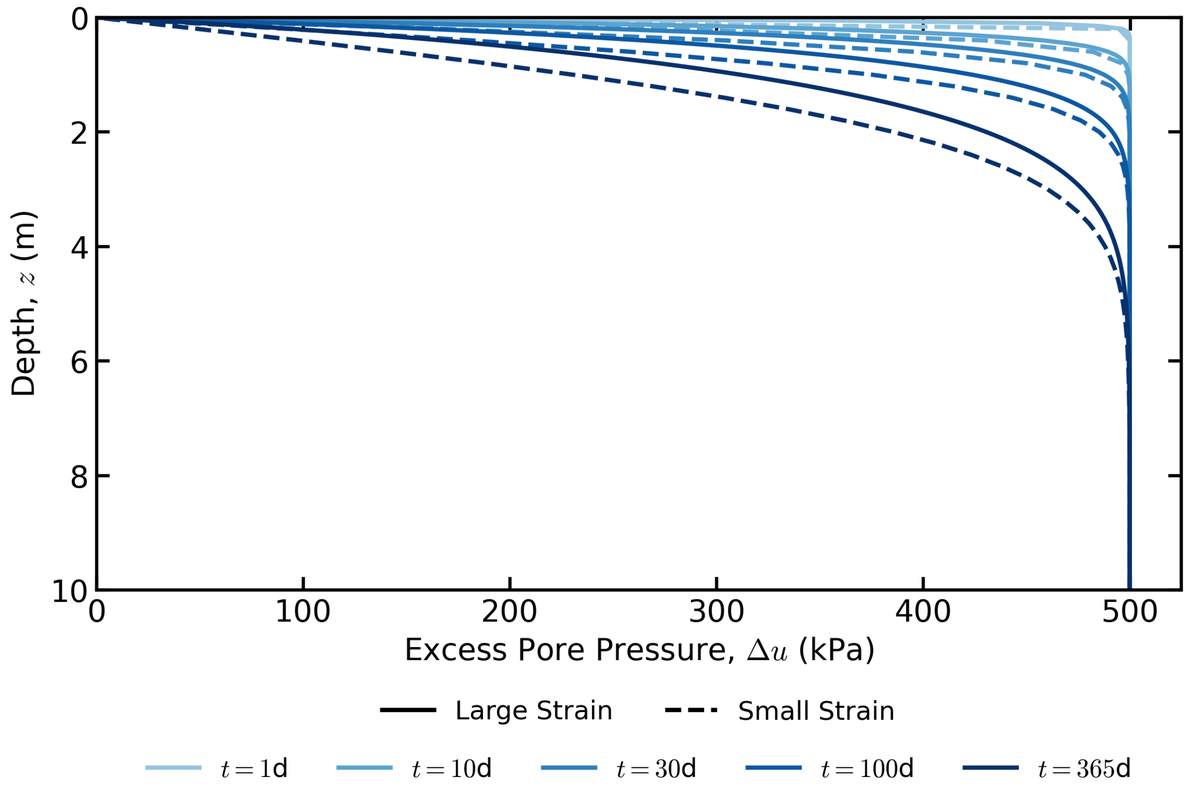
Excellent Agreement

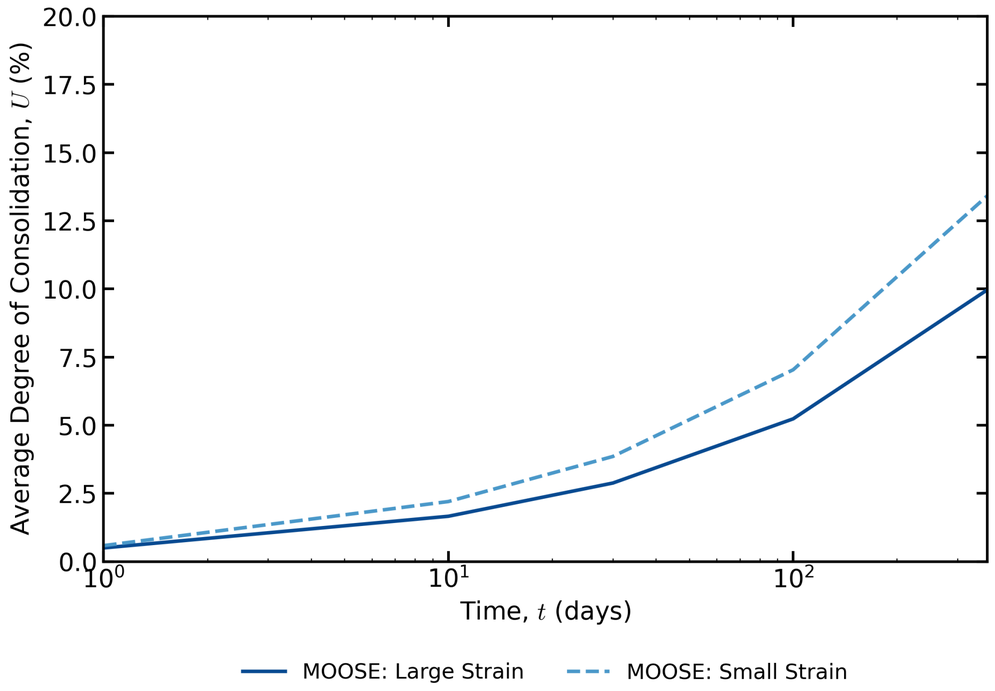
Results for Classical Model (Load = 100 kPa)



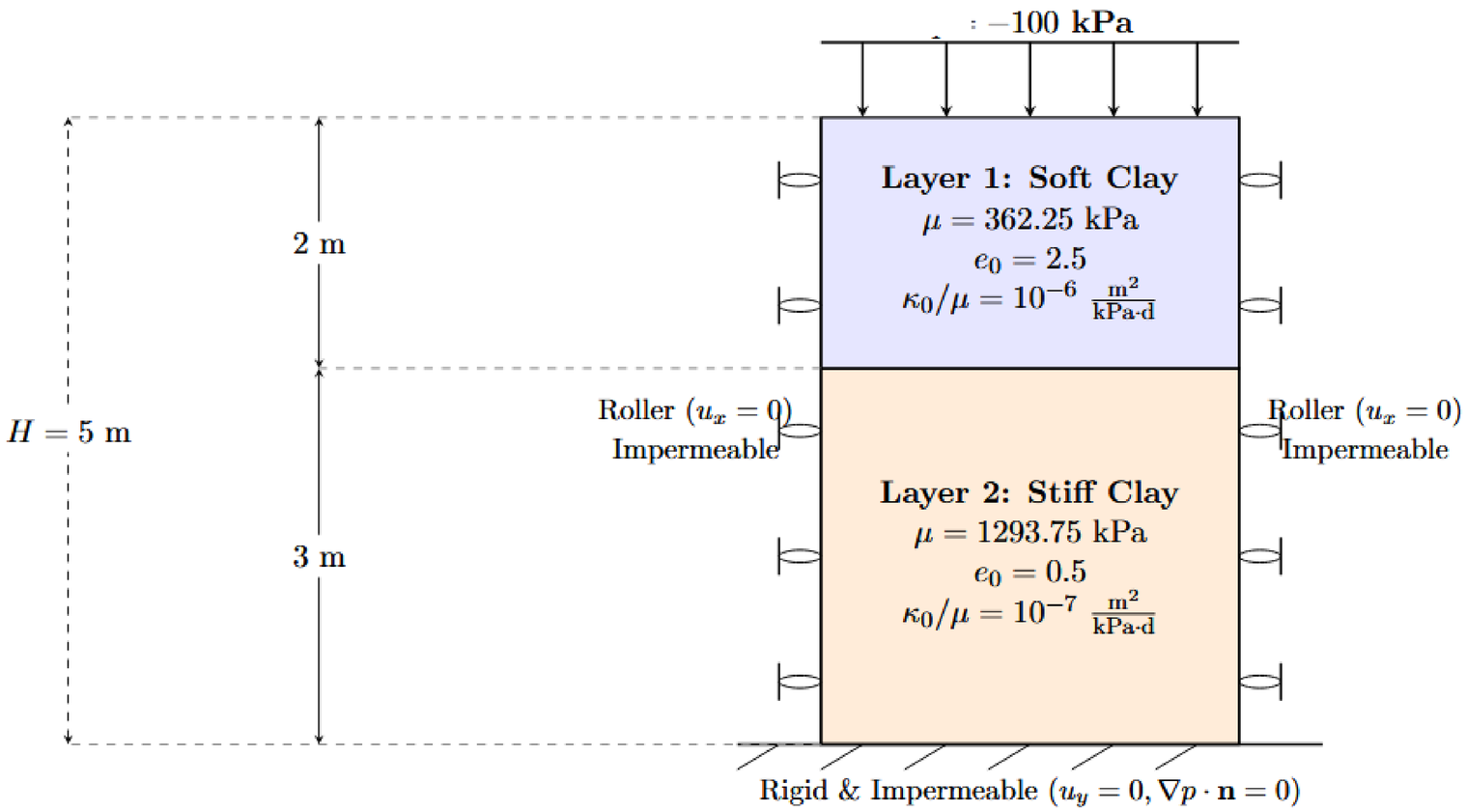


Results for Classical Model (Load = 500 kPa)

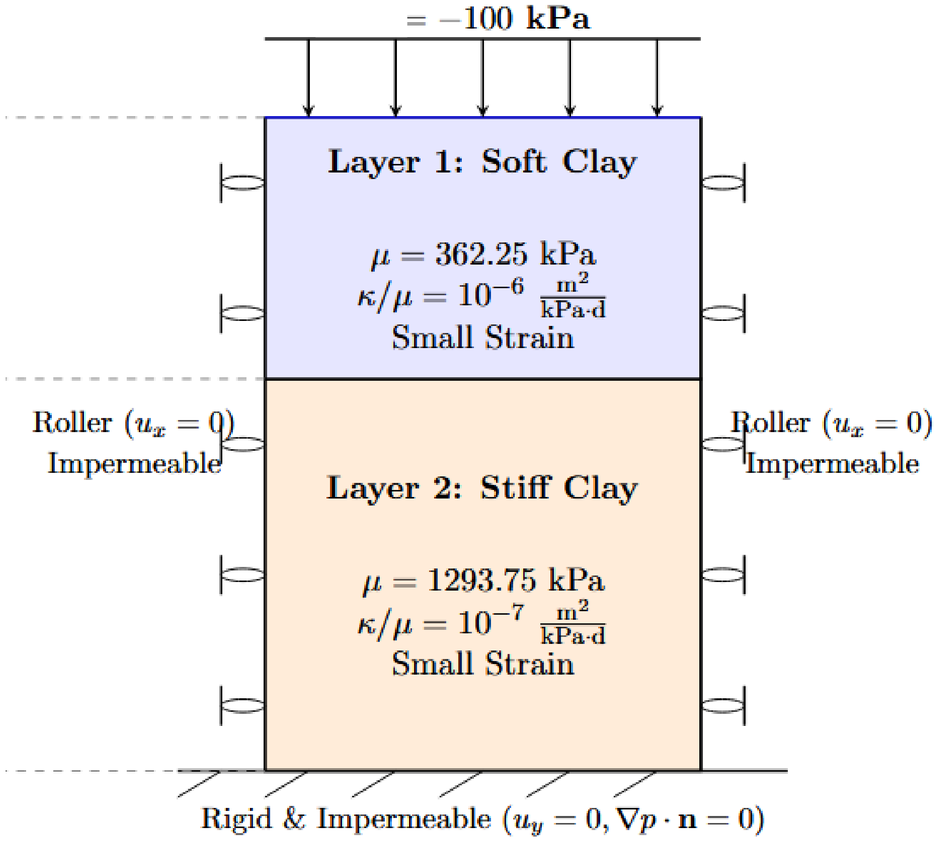




Schematic of the 1D bilayer consolidation model

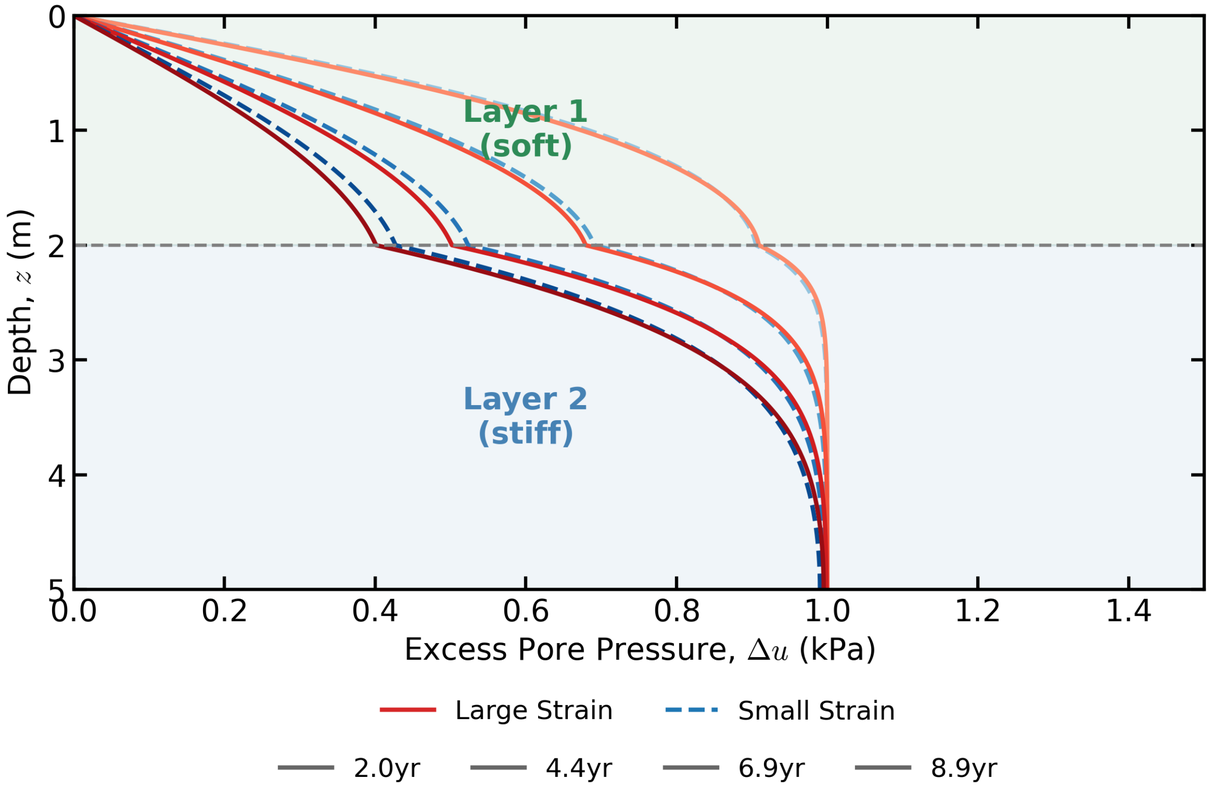


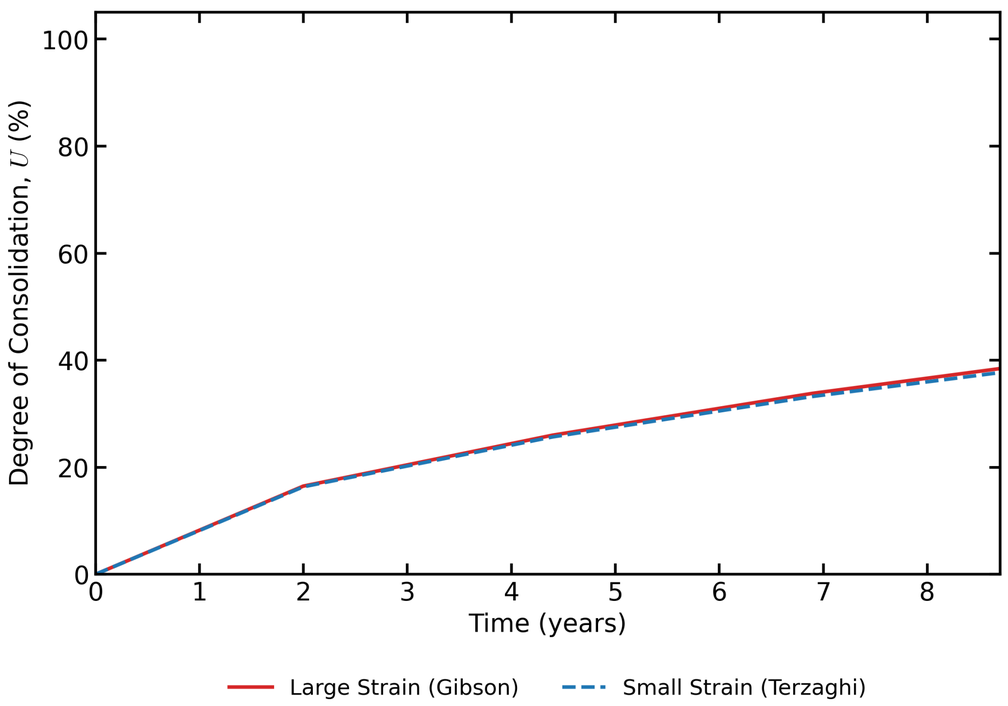
Large strain



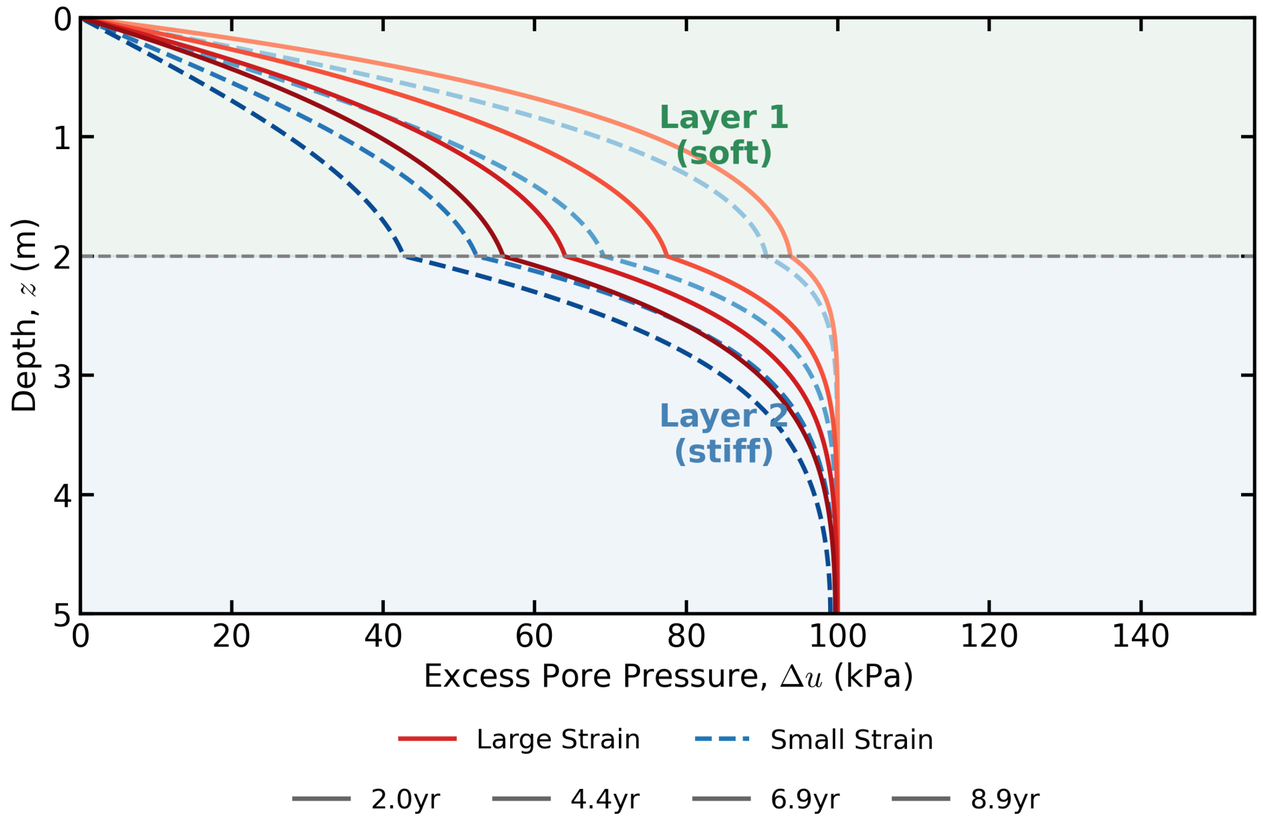
Small strain

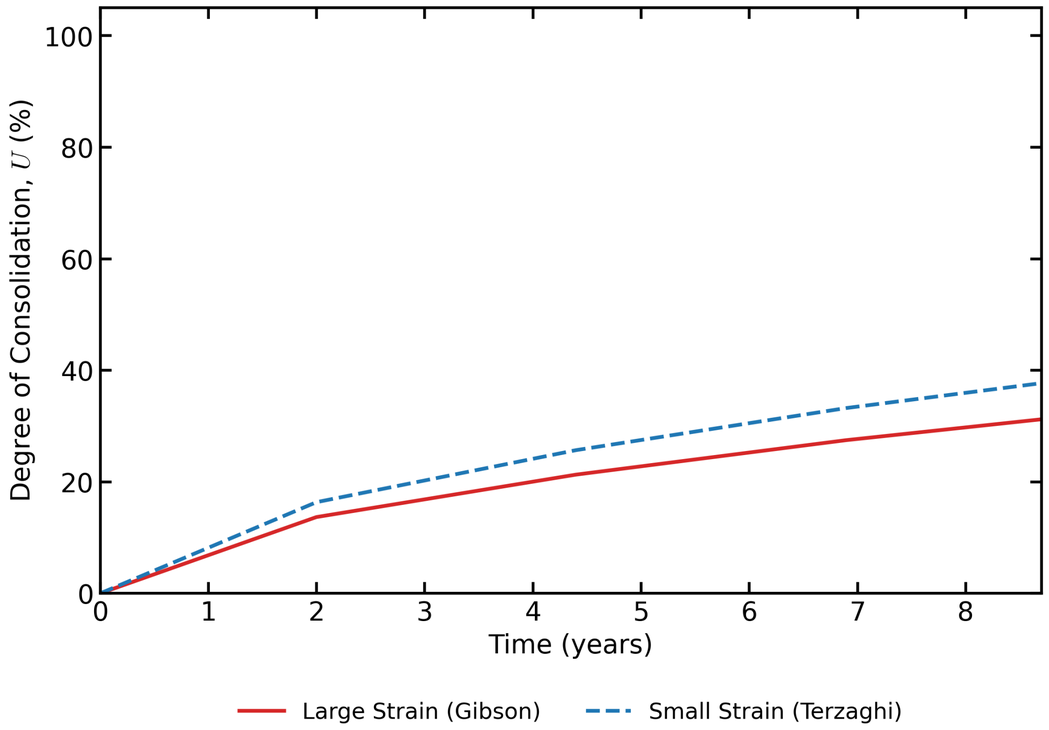
Results for Bi-layer Model (Load = 1 kPa)





Results for Bi-layer Model (Load = 100 kPa)





FINAL NOTES

Discussion

- **Load dependency and non-linearity:** Under minimal loads (1 kPa), small strain and large strain formulations show excellent agreement. However, as applied loads increase to 150 kPa and 500 kPa, small strain theory significantly overpredicts the results.
- **Material sensitivity in complex stratigraphy:** In the bi-layer model, the highly compressible soft clay layer ($e_0 = 2.5$) undergoes geometric changes that violate Cauchy infinitesimal strain assumptions. Capturing this behavior requires the inclusion of quadratic terms via the Green-Lagrange strain formulation.
- **Real-World Implications:** As demonstrated by the Staufen disaster, unpredicted large subsurface volume changes, such as anhydrite swelling, can cause massive uplift and catastrophic structural damage .

Conclusion – Embracing Large Strain Theory

- **Mathematical Necessity:** When soil geometry is virtually unchanged, small strain calculations suffice. However, for significant displacements, excluding quadratic terms leads to a critical divergence in calculated strain and predicted settlement.
- **Robust Multiphysics Implementation:** Transforming governing partial differential equations into their weak form allows for sophisticated numerical solutions . Implementing these within a finite element framework like MOOSE effectively resolves the highly coupled, non-linear mechanics and fluid diffusion inherent in large-strain consolidation .
- **The Engineering Verdict:** Small strain (Terzaghi) theory remains a computationally efficient baseline for stiff materials or negligible loads . However, for soft soils and heavy loading scenarios, large strain (Gibson) theory is not just an academic refinement. It is an engineering imperative to ensure structural safety and accurate predictive modeling.

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Q/A

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